

ABM BUSINESS LOGIC BIBLE

§14 v2 — THE OPPORTUNITY STATE MACHINE & PORTFOLIO ARCHITECTURE

RESOLVING OPEN-Q14.3 — THE BIBLE'S LARGEST ARCHITECTURAL GAP

10 Iterations · Hybrid Path (Option C+) Adopted · 8-State Opportunity Machine · 32 New Rules · Cross-Section Migration Map

§14 v1 surfaced OPEN-Q14.3 honestly rather than inventing a machine no other section used. This pass RESOLVES it. **Account State and Opportunity State are separated into two independent machines.** Account State (§14 v1, unchanged) becomes a Relationship Health summary — the container's lifecycle. Opportunity State (new, this document) becomes the Buying Progress machine — the actual selling lifecycle, scoped to one initiative at one account, with an account able to hold many simultaneously. This is the single most consequential architectural decision since the original Opportunity re-anchoring in §2.

Iterations	10	Opportunity states	8
New rules	32	Sections needing migration	9 (mapped, not yet edited)

THE DECISION — HYBRID PATH (OPTION C+), NOT A OR B

Full machine now (A) creates enormous immediate rework across nine sections. Pure deferral (B) creates compounding technical debt that gets more expensive to fix the longer it's deferred. The adopted path: **(1)** keep §14 v1's Account state machine completely unchanged; **(2)** introduce the Opportunity entity's state machine NOW, fully specified, in this document; **(3)** let downstream modules (§5–§13) migrate to opportunity-scoped operation on their own schedule, guided by the migration map at the end of this document — not rewritten here. The decision is made now; the implementation is sequenced explicitly.

PART A — STATE GOVERNANCE (ITERATIONS 1–4)

Before designing the Opportunity machine, four governance gaps in the EXISTING Account machine needed closing — they generalize beyond Account state to apply to Opportunity state as well, so they're specified once, here, and bind both machines.

LAYER 1 — STATE OWNERSHIP (GOV)

Rule ID	Phase	Specification
STATE-GOV-01	P1	§14 v1's STATE-ACCT-07 (EXEC Stage 5 sole writer of HOT-bound transitions) is GENERALIZED: every state transition in EITHER machine has exactly one declared Authoritative Writer . Examples: WATCHING→ENRICHING = EXEC Stage 5; HOT→OUTREACH_ACTIVE = §9 Outreach Orchestrator; ENGAGED→MEETING_BOOKED = Human + CRM (joint, per §11 v2 HAND-ACC-01's Decision Ownership Boundary); OPPORTUNITY→CUSTOMER = §12 v2's CRM Authority (INT-CRM-02). No transition exists without a named writer; this registers into the §12 v2 Authority Registry (CRM-OWN-03) as the state-transition instance of that general mechanism.
STATE-GOV-02	P1	A state has MANY readers, exactly ONE writer — never multiple writers for the same transition. Where a transition appears jointly owned (e.g. ENGAGED→MEETING_BOOKED), it is specified as a single joint-writer pairing with an explicit protocol (human confirms, system records), not as two independent writers that could race.

LAYER 2 — TRANSITION GUARDS (GUARD)

Rule ID	Phase	Specification
STATE-GUARD-01	P1	Every transition has explicit Entry Conditions beyond 'is this destination reachable from here.' Example: HOT→OUTREACH_ACTIVE requires enrichment complete (§6 ENR-STOP-01 stopping condition met) AND ≥1 Response-Reachable contact (§6 v2 ENR-REACH-04) AND an Agent-5-approved strategy brief exists AND QC is not in a blocked state.
STATE-GUARD-02	P1	A transition request that fails its Entry Conditions creates a Transition Rejection Event (logged, not silently dropped) — retained for §13 v2 analytics (a recurring rejection pattern on the same transition is itself a Failure Cluster candidate, §8 v2 QC-SYS-01).
STATE-GUARD-03	P1	State machine validation checks BOTH destination validity (is this an allowed edge) AND readiness (do entry conditions hold) — §14 v1 only specified the former; this closes the gap.

LAYER 3 — STATE AGING (TIME)

Rule ID	Phase	Specification
STATE-TIME-01	P1	Every state in both machines has an Expected Residence Time : ENRICHING ≤ 48 hours; OUTREACH_ACTIVE ≤ 30 days (matches the 4-touch/14-day sequence plus buffer); HOT_UNENRICHED ≤ 14 days; (Opportunity states get their own residence times, specified in Part B).
STATE-TIME-02	P1	Breaching Expected Residence Time triggers a State Health Alert — NOT an automatic transition. An account stuck in HOT_UNENRICHED for 20 days is flagged for human review, never auto-demoted or auto-retried indefinitely without visibility.
STATE-TIME-03	P2	State Aging (current residence time vs Expected Residence Time, per entity) becomes a standing §13 v2 operational KPI — a system with many entities aging past expectation in the same state is itself a signal worth the same Failure Cluster treatment as a QC anomaly.

LAYER 4 — DERIVED vs STORED STATE (DER)

Rule ID	Phase	Specification
STATE-DER-01	P1	Every state field is classified Authoritative (stored, per §12 v2's Authority Registry) or Derived (computed on read from other authoritative fields, never independently stored). Example: 'does this account have any live Opportunity' is DERIVED from the Opportunity Portfolio (Part B), never a separately-stored boolean that could drift out of sync with the portfolio it should reflect.
STATE-DER-02	P1	Values reliably derivable from already-authoritative fields are NEVER additionally stored — this directly reduces §12 v2's sync-conflict surface (CRM-OWN-01's Authority Registry already requires exactly one owner per field; a derived field's 'owner' is simply the computation itself, with no write path to conflict).
STATE-DER-03	P1	Every entity's field-by-field Authoritative/Derived classification is documented explicitly at the §2 entity-model level, not left implicit — critical for §12 CRM sync, since only Authoritative fields are sync candidates at all (a Derived field is recomputed locally by whichever system needs it, never pushed/pulled).

PART B — THE OPPORTUNITY STATE MACHINE (ITERATIONS 6–10)

The core resolution. Opportunity State models BUYING PROGRESS — a fundamentally different dimension from Account State's RELATIONSHIP LIFECYCLE. They are two parallel machines, never one, and an account's Opportunity Portfolio can contain several Opportunities in different states simultaneously.

LAYER 5 — OPPORTUNITY BIRTH (LIFE)

Rule ID	Phase	Specification
OPP-LIFE-01	P1	An Opportunity is created only when ALL THREE conditions hold simultaneously: (1) an initiative is identified (a named, sourced strategic priority — reuses §6 v2 ENR-INIT-DEEP-01's Public Strategic Initiatives layer as the evidence source); (2) a plausible Decimal solution is mapped to it (a named capability, not a generic 'we might be relevant'); (3) economic value is believed possible (a non-zero Effective Opportunity estimate per §5 v3's existing scoring, not merely a hunch). A signal alone never satisfies this test.
OPP-LIFE-02	P1	Signals (§3 INT-SIG) create HYPOTHESES, not Opportunities. A SIG-HYP hypothesis (§3 reasoning streams) may mature into an Opportunity once OPP-LIFE-01's three conditions are met, but the hypothesis and the Opportunity are explicitly different entities at different confidence levels — this preserves EPIS-RCM-03's anti-false-confidence discipline at the Opportunity-creation boundary specifically.
OPP-LIFE-03	P1	Opportunity creation is an EXPLICIT, logged event (creator, timestamp, the three satisfied conditions cited) — never silently inferred by a background process. This is itself a new Authoritative-Writer transition (Layer 1): Opportunity creation's Authoritative Writer is §5/§6's Opportunity Graph Engine (M19, §2/§3), not any single upstream signal-detection module.

LAYER 6 — OPPORTUNITY IDENTITY & DEDUPLICATION (ID)

Rule ID	Phase	Specification
OPP-ID-01	P1	Every Opportunity has an explicit Opportunity Thesis: {Account, Initiative, Business Outcome} — e.g. 'SAB + Open Banking + API Monetization.' The thesis is the Opportunity's identity, not its account_id alone (which would conflate every initiative at one bank into one Opportunity, defeating the entire re-anchoring).
OPP-ID-02	P1	A new signal attaches to an EXISTING Opportunity when its Initiative-overlap similarity exceeds a configured threshold against an existing open Opportunity's thesis — reusing the SAME dedup-similarity discipline already established for signals (§3 SIG-DEDUP-001's matching logic), applied at the Opportunity-thesis level rather than the individual-signal level. Below threshold, a new Opportunity is created (subject to OPP-LIFE-01).
OPP-ID-03	P1	Opportunity creation triggers a duplication check against all open Opportunities at the same account BEFORE creation completes — a likely-duplicate match (above the OPP-ID-02 threshold but below auto-merge confidence) routes to human review rather than auto-creating or auto-merging silently.

LAYER 7 — OPPORTUNITY DEATH & CLOSURE TAXONOMY (END)

Rule ID	Phase	Specification
OPP-END-01	P1	Opportunity closure requires a mandatory, typed Closure Reason — never a bare WON/LOST binary: WON, LOST_COMPETITOR, LOST_BUDGET, LOST_PRIORITY, LOST_TIMING, MERGED (absorbed into another Opportunity per OPP-ID-02), ABANDONED (no clear reason, sequence simply went cold), PAUSED (not a closure — see Part B state table).
OPP-END-02	P1	§13 v1/v2 learning loops consume the typed Closure Reason, not a binary outcome — LOST_BUDGET and LOST_COMPETITOR imply completely different corrective actions (timing/patience tuning vs. competitive-positioning tuning, §9 v2 Layer 8) and must never be collapsed into one 'LOST' bucket for learning purposes.
OPP-END-03	P1	Closed (WON/LOST_*) Opportunities remain permanently searchable (append-only, per the Bible's standing discipline) — a LOST_TIMING or LOST_BUDGET Opportunity can resurrect into a NEW Opportunity later (new thesis, new creation event per OPP-LIFE-03) with the prior closed record retained as Tier 6 historical context, not deleted or hidden.

LAYER 8 — THE OPPORTUNITY PORTFOLIO (COMP)

This is where Account State stops trying to do two jobs at once. An account with Opportunity A in PROPOSAL and Opportunity B just IDENTIFIED cannot be honestly summarized as a single 'most advanced opportunity' value — doing so hides Opportunity B entirely, exactly the failure mode §14 v1's stopgap (STATE-OPP-02) was honest about being temporary.

Rule ID	Phase	Specification
OPP-COMP-01	P1	Every account maintains an Opportunity Portfolio — the full set of its Opportunities (open and closed), each independently in its own Opportunity State. This REPLACES §14 v1's STATE-OPP-02 stopgap convention (account_state representing 'the most advanced live Opportunity') as the authoritative model.
OPP-COMP-02	P1	Account-level reporting uses a Portfolio Summary (e.g. '3 Active, 1 Proposal, 1 Lost') instead of a single most-advanced-Opportunity proxy — §13 v2 dashboards and §11 v2 handoff packages read the Portfolio Summary, not a collapsed single value.
OPP-COMP-03	P1	Account State (§14 v1, unchanged in its state list) is formally reframed in MEANING, not states: it represents Relationship Health — the container's standing with Decimal — never sales progression. account_state values (WATCHING, HOT, ENGAGED, etc.) describe how actively Decimal is engaging the RELATIONSHIP; they do NOT describe how any individual Opportunity is progressing. This is the conceptual core of the whole resolution.

LAYER 9 — THE OPPORTUNITY STATE MACHINE ITSELF (OPP)

State	Meaning	Expected residence	Transitions
OPP_IDENTIFIED	Initiative detected; hypothesis matured into an Opportunity per OPP-LIFE-01	≤14 days	→ OPP_QUALIFYING → PAUSED → LOST_PRIORITY
OPP_QUALIFYING	Research underway; stakeholders mapped (§6 committee discovery); budget plausibility assessed	≤30 days	→ OPP_ACTIVE → PAUSED → LOST_BUDGET/LOST_PRIORITY
OPP_ACTIVE	Meaningful engagement exists; champion or committee access established	≤60 days	→ OPP_EVALUATING → PAUSED → LOST_TIMING/LOST_COMPETITOR
OPP_EVALUATING	Customer actively evaluating vendors (incumbent + Decimal + others)	≤90 days	→ OPP_PROPOSAL → LOST_COMPETITOR → PAUSED
OPP_PROPOSAL	Commercial discussion underway: proposal/RFP/pricing	≤45 days	→ OPP_NEGOTIATION → LOST_*
OPP_NEGOTIATION	Terms, procurement, security review	≤60 days	→ WON → LOST_*
WON	Closed-won (terminal)	—	→ (new Opportunity may be created for upsell, per §2 EDGE pattern)
LOST_* / MERGED / ABANDONED	Closed, typed reason (terminal)	—	Searchable per OPP-END-03; may inspire a new Opportunity later
PAUSED	Budget freeze, sponsor departure, or timing issue — not closed	Reviewed every 30 days	→ resumes at its PRIOR active state → LOST_* (if paused indefinitely)

Rule ID	Phase	Specification
STATE-OPP-06	P1	The Opportunity State Machine (9 states above) exists INDEPENDENTLY of the Account State Machine (§14 v1, 13 states). They are never merged into one field, and no Opportunity state maps 1:1 onto an Account state.
STATE-OPP-07	P1	Multiple Opportunities may exist simultaneously at one account, each independently positioned in the Opportunity State Machine — this is not an edge case to be handled, it is the NORMAL case the machine is designed for, consistent with §2's original re-anchoring rationale.
STATE-OPP-08	P1	Account State (§14 v1) NEVER derives directly from any single Opportunity's state. Where a summary view is needed (§11 v2 handoff, §13 v2 dashboards), it reads the Portfolio Summary (OPP-COMP-02) — an aggregate across ALL the account's Opportunities — never one Opportunity's state treated as a proxy for the account.
STATE-OPP-09	P1	PAUSED is explicitly NOT a closure state — it preserves the Opportunity's PRIOR active state for resumption (OPP-LIFE pattern: a budget freeze doesn't erase the qualifying work already done) and is reviewed on a fixed 30-day cadence rather than left indefinitely open or forced into a premature LOST closure.

CROSS-SECTION MIGRATION MAP — WHAT THIS CHANGES, SEQUENCED

This decision touches nine downstream sections. None are rewritten here — this is explicit guidance for the next pass through each, so the migration is sequenced and visible rather than silently implied by this document alone.

Section	What migrates to Opportunity-scope	Priority
§2 Entity Model	Opportunity (E38) gains its real state field; Portfolio becomes a queryable relationship, not implicit	Immediate — foundational
§5 Tiering	Score and tier can be computed PER OPPORTUNITY, not only per account — a HOT account may have one PROPOSAL-stage and one just-IDENTIFIED Opportunity needing very different attention	High
§6 Enrichment	Committee mapping becomes Opportunity-specific (the Open Banking buying committee may differ from the Digital Lending committee at the same bank)	High
§7 Personalization	Messaging targets Opportunity context (Active Narrative, AI-NARR-01, becomes per-Opportunity, not per-account)	High
§9 Outreach	Sequences attach to an Opportunity, not only a contact/account pair — the same contact may be in two different sequences for two different Opportunities	Medium
§10 Engagement	Signals and engagement events attribute to the specific Opportunity they bear on (§10 v2 ENG-TRUTH-01's belief-correction tagging already names WHICH hypothesis an event updates — extend that to WHICH Opportunity)	Medium
§11 Handoff	The handoff package becomes Opportunity-centric (OPP-COMP-02's Portfolio Summary replaces the single-Opportunity assumption baked into §11 v1/v2)	High
§12 CRM Sync	Maps naturally onto HubSpot's existing Deal object (one Deal per Opportunity is a closer fit than one Deal per account) — likely simplifies §12 v2 rather than complicating it	Medium
§13 Analytics	Win/loss analysis becomes Opportunity-level, consuming OPP-END-01's typed Closure Reason directly rather than the account-level proxy	High

§14 v2 RULE ROLL-UP & PLACEMENT STATUS

Part	Layer	Rules	Cross-section binding
A	1 Gov	STATE-GOV-01..02	§12 v2 Authority Registry (generalizes into it)
A	2 Guard	STATE-GUARD-01..03	§6 ENR-STOP-01, §8 v2 QC-SYS-01
A	3 Time	STATE-TIME-01..03	§13 v2 operational KPIs
A	4 Der	STATE-DER-01..03	§12 v2 sync-conflict reduction, §2 entity documentation
B	5 Life	OPP-LIFE-01..03	§6 v2 Public Initiatives, §5 v3 Effective Opportunity, EPIS-RCM-03, M19
B	6 ID	OPP-ID-01..03	SIG-DEDUP-001 (reused at thesis level)
B	7 End	OPP-END-01..03	§9 v2 Layer 8 (competitive response), Tier 6 historical memory
B	8 Comp	OPP-COMP-01..03	Replaces §14 v1 STATE-OPP-02 stopgap; reframes Account State's MEANING
B	9 Opp machine	STATE-OPP-06..09	The resolution of OPEN-Q14.3 itself

§14 v2 COMPLETE — OPEN-Q14.3 RESOLVED

The Bible's largest architectural gap is closed. Account State and Opportunity State are now two independent machines: Account State (§14 v1, unchanged) measures Relationship Health; Opportunity State (new, 9 states) measures Buying Progress, scoped to one initiative, with an account able to hold many simultaneously via its Opportunity Portfolio. Opportunity birth requires three explicit conditions (never inferred from a signal alone); identity is protected by thesis-based deduplication reusing the Bible's existing signal-dedup discipline; closure carries a typed reason consumed directly by learning loops; and PAUSED preserves resumability rather than forcing a premature death. State governance was generalized beyond the single HOT-transition fix in §14 v1 to every transition in both machines: named writers, entry guards, residence-time aging, and an explicit Authoritative/Derived classification. 32 new rules.

Nine downstream sections are mapped for migration, not rewritten here — the decision is made; the sequencing is explicit and visible, prioritized by where the current single-Opportunity assumption causes the most real distortion today (§5, §6, §7, §11, §13 first).

PLACEMENT MAP UPDATE

§14 status: upgraded from Complete-with-Opens to **Complete** — OPEN-Q14.3 is RESOLVED (Hybrid Path / Option C+ adopted), not merely deferred. OPEN-Q14.1 and OPEN-Q14.2 (waitlist timeout, relationship-monitor re-engagement) remain open, unaffected by this pass. The migration map's nine entries are tracked as a standing cross-section to-do, not blocking §14's own completion.

ON THE HORIZON

§17 KSA-Specific Business Rules remains the Bible's only fully 'Not started' section. §16 Timing Matrix consolidation is the natural lighter companion. Separately, the migration map above is available whenever you want to revisit §5, §6, §7, §11, or §13 specifically to apply Opportunity-scoping to them.